



Advanced Time Series Analysis and Forecasting

Chapter 1: Forecast evaluation, scoring rules and combination



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/3)

- ▣ AI: Artificial Intelligence
- ▣ ALFRED: Archival Federal Reserve Economic Data
- ▣ AO: Atkeson–Ohanian (naive inflation forecast)
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARX: AutoRegressive model with eXogenous variables
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ CPI: Consumer Price Index
- ▣ CPIAUCSL: Consumer Price Index for All Urban Consumers (FRED series code)
- ▣ CRPS: Continuous Ranked Probability Score
- ▣ CW: Clark–West (test for nested models)
- ▣ DM: Diebold–Mariano (test)
- ▣ EA: Euro Area
- ▣ ECB: European Central Bank
- ▣ ENTSO-E: European Network of Transmission System Operators for Electricity
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ ES: Expected Shortfall
- ▣ FRED: Federal Reserve Economic Data
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GARCH-N: GARCH with Normal innovations
- ▣ GDP: Gross Domestic Product
- ▣ GW: Giacomini–White (test of conditional predictive ability)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HICP: Harmonised Index of Consumer Prices
- ▣ HLN: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- ▣ INS: Institutul Național de Statistică (the National Institute of Statistics (Romania))
- ▣ ISE: (Fraunhofer) Institute for Solar Energy Systems

Acronyms used in this material (2/3)

- ▣ LLM: Large Language Model
- ▣ LR: Likelihood Ratio (test)
- ▣ MA: Moving Average
- ▣ MAE: Mean Absolute Error
- ▣ MAPE: Mean Absolute Percentage Error
- ▣ MASE: Mean Absolute Scaled Error
- ▣ MCS: Model Confidence Set
- ▣ MSE: Mean Squared Error
- ▣ MSPE: Mean Squared Prediction Error
- ▣ MW: MegaWatt
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ OLS: Ordinary Least Squares
- ▣ PC: Phillips Curve
- ▣ PIT: Probability Integral Transform
- ▣ QLIKE: Quasi-LIKelihood loss
- ▣ RC: White's Reality Check (for data snooping)
- ▣ RMSE: Root Mean Squared Error
- ▣ RTDSM: Real-Time Data Set for Macroeconomists (Philadelphia Fed)
- ▣ RW: Random Walk
- ▣ SE: Standard Error
- ▣ SPA: Superior Predictive Ability (test)
- ▣ SPF: Survey of Professional Forecasters (Federal Reserve Bank of Philadelphia)
- ▣ TSA: Time Series Analysis
- ▣ UIP: Uncovered Interest Parity
- ▣ UNRATE: Unemployment Rate (FRED series code)
- ▣ US: United States

Acronyms used in this material (3/3)

- ▣ **VAR: Vector AutoRegression**

Today's question and route

- ▣ **Question:** when is one forecast *really* better than another, and what do we gain by combining them?
 - ▶ “better” needs a loss or a score, a reference distribution of the comparison and an honest evaluation design
- ▣ **Route** of the chapter
 - ▶ point forecasts: which functional a loss function rewards (consistency, Bregman losses)
 - ▶ probabilistic forecasts: calibration, sharpness, PIT, proper scoring rules, elicibility
 - ▶ comparing forecasts: DM with HAC and small-sample corrections, GW, Clark–West, MZ, encompassing, RC, SPA, MCS
 - ▶ combination: Bates–Granger, the combination puzzle, density pools, the M4 and M5 lessons; real-time data
- ▣ We build on [TSA, Chapter 0](#) and [TSA, Chapter 4](#) (MAE, RMSE, MASE, rolling-origin evaluation, the basic DM test) and on [Chapter 0](#) (HAC, bootstrap); Seminar 1 comes before this lecture

Learning outcomes

- ▣ Derive the optimal point forecast under squared, absolute, pinball and asymmetric losses, and recognise the losses consistent for the mean (Bregman)
- ▣ Assess the calibration of density forecasts with PIT histograms and tests, and rank them with proper scores (log score, CRPS, interval and energy scores)
- ▣ Explain elicibility and why VaR is elicitable while ES is elicitable only jointly with VaR
- ▣ Test equal predictive ability correctly: DM–HLN with HAC variance, GW, Clark–West for nested models, MZ, encompassing, SPA and the MCS
- ▣ Combine point and density forecasts and explain the forecast combination puzzle; evaluate forecasts with real-time data

Reading, data and tools

- ▣ Surveys: [Petropoulos et al. \(2022\)](#) (open access), [Gneiting and Katzfuss \(2014\)](#), [Elliott and Timmermann \(2008\)](#), [Timmermann \(2006\)](#), [Wang et al. \(2023\)](#)
 - ▶ textbooks: [Hamilton \(1994\)](#), Ch. 4 (forecasting); [Kilian and Lütkepohl \(2017\)](#); bridge: [Huang and Petukhina \(2022\)](#)
- ▣ Python Quantlets of this chapter: [Quantlets/Ch_01](#)
 - ▶ DM–HLN, GW, Clark–West, MZ, encompassing, Berkowitz, CRPS and the MCS written out in `numpy`; Hansen’s SPA from the `arch` package
- ▣ Lecture notebook: [open in Google Colab](#)

Data used in this chapter

Series	Source	Use
HICP inflation, Romania (y/y, monthly)	Eurostat (prc_hicp_minr)	DM, GW
CPI and unemployment, US (quarterly)	FRED (CPIAUCSL, UNRATE)	Atkeson–Ohanian
EUR/RON and 3-month rates RO, EA (monthly)	BNR reference rate; Eurostat (irt_st_m)	Clark–West, SPA
S&P 500, daily close	EODHD	PIT, scores
Electricity load, Romania (hourly)	Energy-Charts (Fraunhofer ISE), ENTSO-E data	pinball, MCS
Individual forecasts, US SPF	Federal Reserve Bank of Philadelphia	combination
US real GDP, all releases	Philadelphia Fed, RTDSM	real time

- All sources are public and need no account or key; daily and monthly data up to September 2026
- HICP: harmonised index of consumer prices; CPI: consumer price index; SPF: Survey of Professional Forecasters; RTDSM: Real-Time Data Set for Macroeconomists

From weather forecasts to forecast verification



Robert FitzRoy (1805–1865)

Portrait: Charles Hemus (c. 1910); public domain;

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Lewis Fry Richardson (1881–1953)

Photo: NOAA; public domain; Wikimedia Commons

- 1861: FitzRoy, head of the new Meteorological Office, publishes daily “forecasts” in The Times
 - ▶ he coined the word in its modern sense; the forecasts were often wrong and publicly mocked
- 1922: Richardson computes a weather forecast by numerical integration of the equations of motion, by hand
- 1950: Brier (1950) proposes a score for probability forecasts of rain; meteorology leads forecast verification for the next fifty years
 - ▶ economics adopts the same tools late: PIT Diebold, Gunther and Tay (1998), proper scores Gneiting and Raftery (2007), calibration Gneiting, Balabdaoui and Raftery (2007)
- Lesson: a forecast is only as useful as the way we evaluate it

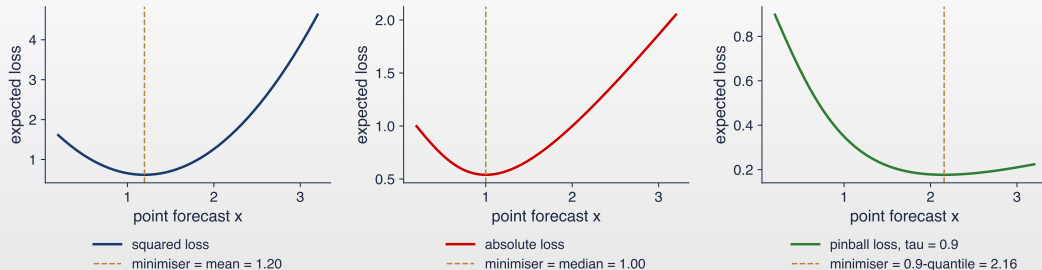
Setting and notation

- Target Y_{t+h} , information set $\mathcal{F}_t$, horizon h ; a **point forecast** $\hat{y}_{t+h|t}$ is $\mathcal{F}_t$ -measurable
 - ▶ a **probabilistic forecast** is a predictive distribution $F_{t+h|t}$ (density, quantiles, interval, ensemble)
- Out-of-sample design: R observations to estimate, P forecasts to evaluate, $R + P + h - 1 = T$
 - ▶ schemes: **fixed** (estimate once), **rolling** (last R), **recursive** (all past); TSA, Chapter 4 called this rolling-origin evaluation
 - ▶ the scheme matters for inference: West (1996) vs Giacomini–White (2006), Section 4
- Known from TSA, Chapter 0: MAE, RMSE, MAPE and the scale-free MASE Hyndman and Koehler (2006)
 - ▶ here the question is different: *which* loss, and what does it reward?

The optimal point forecast and consistency

- Given a loss $L(x, y)$ and a predictive distribution F , the **optimal** (Bayes) forecast is $x^* = \arg \min_x \mathbb{E}_F[L(x, Y)]$
 - ▶ a loss is **consistent** for a functional T if $\mathbb{E}_F L(T(F), Y) \leq \mathbb{E}_F L(x, Y)$ for all x and F [Gneiting \(2011\)](#)
 - ▶ **strictly** consistent: equality only at $x = T(F)$
- Three classical cases ($e = y - x$)
 - ▶ squared e^2 : $\frac{d}{dx} \mathbb{E}(Y - x)^2 = -2\mathbb{E}(Y - x) = 0 \Rightarrow x^* = \mathbb{E}Y$ (the mean)
 - ▶ absolute $|e|$: $\frac{d}{dx} \mathbb{E}|Y - x| = F(x) - (1 - F(x)) = 0 \Rightarrow F(x^*) = 1/2$ (the median)
 - ▶ pinball $\rho_\tau(e) = (\tau - 1\{e < 0\})e$: $F(x) - \tau = 0 \Rightarrow x^* = F^{-1}(\tau)$ (the τ -quantile)
- Consequence: report the loss *before* asking for a forecast; a median forecast judged by RMSE is judged unfairly [Gneiting \(2011\)](#)

Three losses, three optimal forecasts



□ Y log-normal with $\ln Y \sim N(0, 0.6^2)$; expected loss as a function of the point forecast x (simulation, 2×10^5 draws)

ATS_ch1_loss_functions

Interpreting the three minimisers

- Mean 1.20, median 1, 0.9-quantile 2.16: three different “best” forecasts of the same variable
 - ▶ for a right-skewed target (wages, loads, prices, volatility) the gap is large
- The pinball curve is flat near its minimum: quantile forecasts are hard to separate with few observations
- A forecaster who knows the loss reports the matching functional; ranking forecasts by another loss rewards the wrong behaviour

Bregman losses: all losses consistent for the mean

- **Theorem Savage (1971), Gneiting (2011)**: under regularity, L is consistent for the mean iff
 - ▶ $L(x, y) = \phi(y) - \phi(x) - \phi'(x)(y - x)$ with ϕ convex (a **Bregman** loss), up to terms in y only
 - ▶ $\phi(x) = x^2$: squared error; $\phi(x) = -\ln x$: **QLIKE** $y/x - \ln(y/x) - 1$ ([Chapter 8](#)); the conditional mean is optimal for every Bregman loss [Banerjee et al. \(2005\)](#); proof for the quantile in the [Appendix](#)
- Proof sketch: $\mathbb{E}L(x, Y) - \mathbb{E}L(\mu, Y) = \phi(\mu) - \phi(x) - \phi'(x)(\mu - x) \geq 0$ by convexity
 - ▶ the linear term vanishes because $\mathbb{E}(Y - \mu) = 0$
- Correctly specified forecasts rank the same under every Bregman loss; misspecified ones need not [Patton \(2020\)](#)
 - ▶ Murphy diagrams [Ehm et al. \(2016\)](#) plot the score over the whole family: one forecast dominates only if its curve is lower everywhere
 - ▶ robust proxies for volatility: only some losses keep the ranking with a noisy proxy [Patton \(2011\)](#)

Asymmetric loss

- **LinEx** loss $L(e) = b[\exp(ae) - ae - 1]$, $e = y - x$: $a > 0$ penalises under-forecasts exponentially
 - ▶ optimal forecast $x^* = a^{-1} \ln \mathbb{E} e^{aY}$; with $Y | \mathcal{F}_t \sim N(\mu_t, \sigma_t^2)$: $x^* = \mu_t + a\sigma_t^2/2$ [Christoffersen and Diebold \(1997\)](#)
 - ▶ the optimal forecast is **biased** and its bias moves with the conditional variance
- Consequences for evaluation
 - ▶ a rational forecaster under asymmetric loss fails the MZ unbiasedness test (Section 4)
 - ▶ the pinball loss is the piecewise-linear member: the optimal forecast is a quantile
- Examples: a grid operator (under-forecasting load is costlier than over-forecasting), a central bank with asymmetric inflation costs, the capital buffer of a bank ([Chapter 9](#))

Recap: Point forecasts and loss functions

- A loss defines the target functional: squared $\rightarrow$ mean, absolute $\rightarrow$ median, pinball $\rightarrow$ quantile
- Bregman losses are exactly those consistent for the mean; QLIKE is one of them
- Under asymmetric loss the optimal forecast is biased by design
- With misspecified models the ranking can depend on the loss: fix it in advance

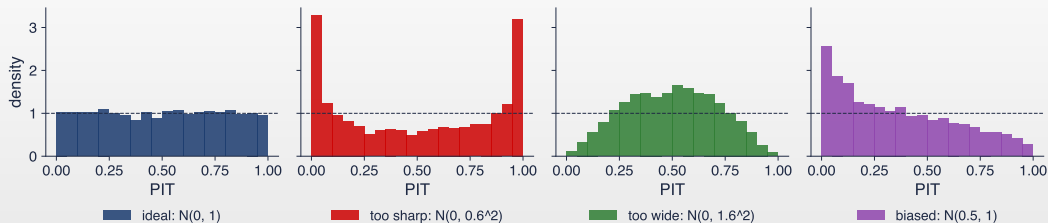
Maximise sharpness subject to calibration

- ▣ **Calibration:** statistical consistency between the forecasts F_t and the outcomes y_t (a joint property) [Gneiting, Balabdaoui and Raftery \(2007\)](#)
 - ▶ **probabilistic** calibration: the PIT $u_t = F_t(y_t)$ is uniform on average
 - ▶ **marginal** calibration: $\frac{1}{T} \sum_t F_t(x) \rightarrow G(x)$, the limit of the empirical distribution of y_t
- ▣ **Sharpness:** concentration of the predictive distributions; a property of the forecasts only
 - ▶ measured by the average width of central intervals or the average predictive variance
- ▣ Paradigm [Gneiting, Balabdaoui and Raftery \(2007\)](#): among calibrated forecasts, prefer the sharpest
 - ▶ the climatological forecast (the unconditional distribution) is calibrated but not sharp
 - ▶ proper scores (next section) reward both at once

The probability integral transform

- **Theorem** Dawid (1984), Diebold, Gunther and Tay (1998): if F_t is the true conditional distribution of Y_t given $\mathcal{F}_{t-1}$ and continuous, then $u_t = F_t(y_t)$ are i.i.d. $U(0, 1)$
 - ▶ uniform: $P(u_t \leq v \mid \mathcal{F}_{t-1}) = P(Y_t \leq F_t^{-1}(v) \mid \mathcal{F}_{t-1}) = v$ for every past
 - ▶ independent: a conditional distribution that does not depend on the past implies independence of the sequence
- For $h > 1$ the PITs are uniform but serially dependent up to lag $h - 1$ (overlapping horizons)
 - ▶ tests must allow this dependence Rossi and Sekhposyan (2019)
- Uniformity is necessary, not sufficient: Hamill (2001) and Gneiting, Balabdaoui and Raftery (2007) build uncalibrated forecasters whose PIT is uniform on average
 - ▶ check PIT conditionally: by sub-sample, by regime, by horizon

What PIT histograms reveal



□ $Y \sim N(0, 1)$, 5000 draws; four Normal forecasts; dashed line: the uniform density

ATS_ch1_calibration_scores

Interpreting the PIT shapes

- ▣ **U-shape**: too many outcomes in both tails (32% outside the central 90%, against 10%): the forecast is too sharp, intervals too narrow
- ▣ **Hump**: too few outcomes in the tails (1.0%): the forecast is too wide, the intervals waste information
- ▣ **Slope**: a biased location; here the forecast mean is too high, so PIT values cluster near 0
- ▣ CRPS (lower is better): ideal 0.567, too sharp 0.595, too wide 0.604, biased 0.637: the proper score ranks the ideal forecast first

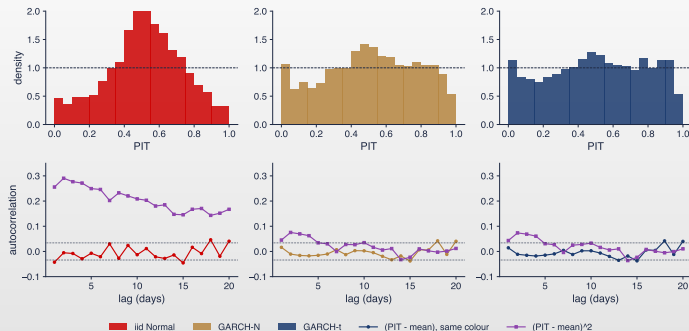
Testing calibration

- ▣ **Berkowitz (2001)**: $z_t = \Phi^{-1}(u_t)$ should be i.i.d. $N(0, 1)$ [Berkowitz \(2001\)](#)
 - ▶ fit $z_t = \mu + \rho(z_{t-1} - \mu) + \varepsilon_t$, $\varepsilon_t \sim N(0, \sigma^2(1 - \rho^2))$; LR test of $\mu = 0, \sigma = 1, \rho = 0$, $\chi^2(3)$
 - ▶ the transformation to z makes the tails visible; it tests only the first two moments and AR(1) dependence
- ▣ **Correlograms** of $(u_t - \bar{u})^k$, $k = 1, 2$ [Diebold, Gunther and Tay \(1998\)](#): dependence in level and in volatility
 - ▶ Ljung–Box on $(u_t - \bar{u})^2$ detects missing volatility dynamics
- ▣ Parameter estimation changes the null distribution of PIT tests; with P/R small it can be ignored, otherwise use [Rossi and Sekhposyan \(2019\)](#)
- ▣ Interval forecasts: coverage and independence of the hits [Christoffersen \(1998\)](#); for VaR this becomes backtesting ([Chapter 9](#))

Case study: Diebold, Gunther and Tay (1998)

- The paper: daily S&P 500 returns; parameters estimated once on the first part of the sample, density forecasts evaluated on the second part with PIT histograms and correlograms of $(u - \bar{u})^k$ Diebold, Gunther and Tay (1998)
 - ▶ their models: i.i.d. Normal; MA(1)-GARCH(1,1) with Normal and with Student- t innovations Bollerslev (1987)
- Our replication: the same three models, estimated by maximum likelihood on 2000–2012, fixed parameters, one-day density forecasts for 2013–2026 (3 448 days)
 - ▶ GARCH- t : $\hat{\theta} = -0.061$, $\hat{\alpha} + \hat{\beta} = 0.996$, $\hat{\nu} = 8.0$ degrees of freedom
- Question: which density forecast is calibrated, and do the proper scores agree with the PIT diagnosis?

PIT diagnostics of three S&P 500 density forecasts



Top: PIT histograms (20 bins); bottom: autocorrelations of $(u - \bar{u})$ and $(u - \bar{u})^2$ with $\pm 2/\sqrt{T}$ bands

ATS_ch1_density_forecasts

Interpreting the S&P 500 diagnostics

- i.i.d. Normal: a hump and strong autocorrelation of $(u - \bar{u})^2$: the forecast ignores volatility clustering; Berkowitz LR 428.8 ($p < 0.001$)
 - ▶ too wide in calm periods, too narrow in turbulent ones: the two errors average into a hump
- GARCH removes most of the dependence; the t version has the better left tail
 - ▶ Berkowitz: GARCH-N LR 9.3 (p 0.026), GARCH- t LR 10.0 (p 0.019); estimated σ_z $0.97 < 1$
 - ▶ outside the central 90%: 8.4% of days: with parameters frozen in 2012 the forecasts are slightly too wide after 2013
- In line with [Diebold, Gunther and Tay \(1998\)](#): conditional heteroskedasticity is essential, fat tails help; a small residual miscalibration remains

Recap: Calibration

- ▣ Calibration is a joint property of forecasts and outcomes; sharpness is a property of the forecasts
- ▣ Correct one-step densities give i.i.d. uniform PITs; for $h > 1$ they are dependent
- ▣ U-shape: too sharp; hump: too wide; slope: bias; correlogram of $(u - \bar{u})^2$: missing volatility dynamics
- ▣ The S&P 500: GARCH- t is close to calibrated, the i.i.d. Normal is not

Proper scoring rules

- A **scoring rule** $S(F, y)$ assigns a penalty to the forecast F when y occurs (negatively oriented: lower is better)
 - ▶ **proper**: $\mathbb{E}_G S(G, Y) \leq \mathbb{E}_G S(F, Y)$ for all F, G : reporting the true belief is optimal [Gneiting and Raftery \(2007\)](#)
 - ▶ **strictly proper**: equality only if $F = G$; then the score cannot be gamed
- Proper scores reward calibration and sharpness together: the expected score decomposes into uncertainty, minus resolution, plus reliability
 - ▶ the Brier score $(p - 1\{\text{event}\})^2$ for binary events is the oldest example [Brier \(1950\)](#)
- A proper score of a probabilistic forecast reduces to a consistent loss of a point forecast when F is a point mass (CRPS $\rightarrow$ absolute error)

The logarithmic score

- $\text{LogS}(f, y) = -\ln f(y)$ [Good \(1952\)](#): the only proper score that is **local** (depends on f only at y , up to equivalence)
 - ▶ properness: $\mathbb{E}_g[\text{LogS}(f, Y)] - \mathbb{E}_g[\text{LogS}(g, Y)] = \int g \ln(g/f) = \text{KL}(g\|f) \geq 0$ (Gibbs inequality)
- Links: average log score = minus the out-of-sample log-likelihood; the difference of two log scores is a predictive likelihood ratio
 - ▶ Bayes factors and the predictive model choice of [Geweke and Amisano \(2011\)](#) are built on it
- Weaknesses: infinite if $f(y) = 0$; very sensitive to tail events; needs a density
 - ▶ cannot score ensembles or quantile forecasts directly

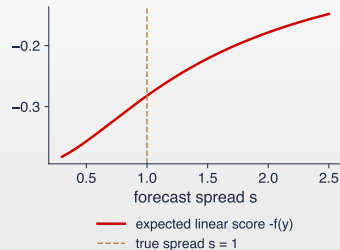
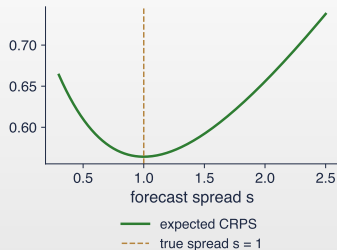
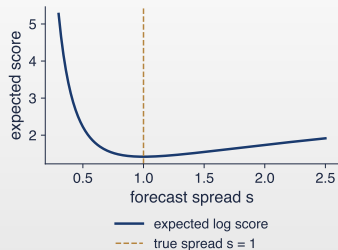
The continuous ranked probability score

- $\text{CRPS}(F, y) = \int_{-\infty}^{\infty} (F(z) - 1\{y \leq z\})^2 dz$ Matheson and Winkler (1976)
 - ▶ kernel form: $\text{CRPS}(F, y) = \mathbb{E}_F |X - y| - \frac{1}{2} \mathbb{E}_F |X - X'|$, $X, X' \sim F$ independent Gneiting and Raftery (2007)
(proof in the [Appendix](#))
 - ▶ first term: accuracy; second: a reward for spread that stops the score from favouring point masses
- Closed forms: $N(\mu, \sigma^2)$: $\sigma [z(2\Phi(z) - 1) + 2\varphi(z) - 1/\sqrt{\pi}]$, $z = (y - \mu)/\sigma$; also for Student t (Quantlet)
 - ▶ ensembles: plug in the empirical distribution of the members
- Decompositions: $\text{CRPS} = \int \text{BS}_z dz = 2 \int_0^1 \rho_\tau(y - F^{-1}(\tau)) d\tau$
 - ▶ the integral of Brier scores over thresholds, or twice the integral of pinball losses over quantile levels
 - ▶ so the average pinball over the 99 percentiles (GEFCom2014) approximates $\text{CRPS}/2$ Hong et al. (2016)

Quantile, interval and multivariate scores

- **Quantile score:** $\rho_\tau(y - q)$, strictly consistent for the τ -quantile; any g increasing gives $(1\{y \leq q\} - \tau)(g(q) - g(y))$ [Gneiting \(2011\)](#)
- **Interval score** of a central $(1 - \alpha)$ interval $[l, u]$ [Gneiting and Raftery \(2007\)](#)
 - ▶ $IS_\alpha = (u - l) + \frac{2}{\alpha}(l - y)1\{y < l\} + \frac{2}{\alpha}(y - u)1\{y > u\}$: width plus a penalty for misses
 - ▶ it equals $\frac{2}{\alpha}[\rho_{\alpha/2}(y - l) + \rho_{1-\alpha/2}(y - u)]$; coverage alone is not a score
- **Energy score** for vectors: $ES(F, y) = \mathbb{E}\|X - y\| - \frac{1}{2}\mathbb{E}\|X - X'\|$, the multivariate CRPS
 - ▶ weak at detecting wrong correlations; the **variogram score** [Scheuerer and Hamill \(2015\)](#) targets the dependence structure (24 hourly loads, a VAR path)
- **Tails:** threshold-weighted CRPS $\int (F(z) - 1\{y \leq z\})^2 w(z) dz$ stays proper [Gneiting and Ranjan \(2011\)](#); the weighted likelihood ratio of [Amisano and Giacomini \(2007\)](#) does not [Gneiting and Ranjan \(2011\)](#)

Proper and improper scores



- Truth $Y \sim N(0, 1)$; forecast $N(0, s^2)$; expected score as a function of the spread s (closed forms)
- Log score and CRPS: minimum at $s = 1$; linear score $-f(y)$: $\mathbb{E}f(Y) = [2\pi(1 + s^2)]^{-1/2}$ grows as $s \rightarrow 0$: it rewards overconfidence

Interpreting the expected scores

- The log score and the CRPS are minimised by the true spread: a forecaster maximises the expected reward by reporting what she believes
- The log score punishes overconfidence much more than excess spread (steep left branch); the CRPS is more symmetric and bounded by the absolute error
- The linear score is maximised by a point mass: it would reward a forecaster who always reports a spike, so it must not be used for evaluation

Elicibility

- A functional T is **elicitable** if a strictly consistent loss exists for it [Gneiting \(2011\)](#)
 - ▶ mean, quantiles, expectiles, ratios of expectations: elicitable
 - ▶ only elicitable functionals can be compared and ranked meaningfully with point forecasts
- **Necessary condition** (Osband): convex level sets: $T(F_0) = T(F_1) = t \Rightarrow T(\lambda F_0 + (1 - \lambda)F_1) = t$
 - ▶ the **variance** fails it, so it is not elicitable; the pair (mean, second moment) is
- Risk measures with $\text{VaR}_\alpha(X) = -q_\alpha(X)$ (VaR 1%: $\alpha = 0.01$)
 - ▶ VaR is a quantile: elicitable by the pinball loss
 - ▶ ES has non-convex level sets: **not elicitable** alone [Weber \(2006\)](#), [Gneiting \(2011\)](#); (VaR, ES) is **jointly** elicitable [Fissler and Ziegel \(2016\)](#)
 - ▶ the Fissler–Ziegel scores and the backtests built on them: [Chapter 9](#)

Scores of the S&P 500 density forecasts

Model	Log score	CRPS	Outside 90%	Berkowitz p
i.i.d. Normal	1.531	0.564	3.9%	< 0.001
MA(1)-GARCH(1,1)-N	1.265	0.509	8.0%	0.026
MA(1)-GARCH(1,1)- t	1.233	0.507	8.4%	0.019

- Score differences tested with DM on the daily scores ([Amisano and Giacomini \(2007\)](#), unweighted)
 - GARCH- t against GARCH-N: HLN -5.9 (log score), -6.4 (CRPS); GARCH-N against i.i.d.: -12.7 and -21.5
- Both scores rank the models in the same order as the PIT diagnosis; the log score separates the tails more sharply
- Ranking and calibration are different questions: the best-scoring model can still fail a calibration test

Recap: Proper scores and elicibility

- Proper scores make honesty optimal; strictly proper scores cannot be gamed
- LogS: local, tail-sensitive; CRPS: robust, works for ensembles, equals twice the integrated pinball loss
- Interval score for intervals, energy and variogram scores for vectors
- Elicitable: mean, quantile (VaR); not elicitable: variance, ES alone; (VaR, ES) jointly

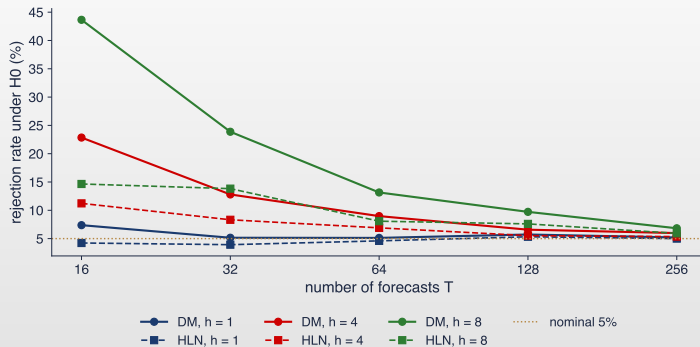
Two questions about predictive ability

- ▣ **Population** (West 1996): are the *models*, at their pseudo-true parameters, equally accurate? [West \(1996\)](#)
 - ▶ estimation error enters the asymptotic variance unless $P/R \rightarrow 0$ or the loss is the one used in estimation
- ▣ **Finite sample** (Giacomini–White 2006): are the *forecasting methods*, with their estimation windows, equally accurate? [Giacomini and White \(2006\)](#)
 - ▶ rolling window of fixed size R : estimation noise is part of the method; nested models allowed
- ▣ **Diebold–Mariano** [Diebold and Mariano \(1995\)](#) treats the forecasts as given (“primitives”): it compares forecasts, not models [Diebold \(2015\)](#)
 - ▶ using DM to compare models with estimated parameters needs the West or GW framework

The Diebold–Mariano test with HAC variance

- Loss differential $d_t = L(e_{1t}) - L(e_{2t})$, $H_0: \mathbb{E}d_t = 0$; assume d_t covariance stationary
 - ▶ $DM = \bar{d} / \sqrt{\hat{\sigma}_{LR}^2 / P} \rightarrow N(0, 1)$, $\sigma_{LR}^2 = \gamma_0 + 2 \sum_{k \geq 1} \gamma_k$ the long-run variance ([Chapter 0](#))
- Optimal h -step errors are at most $MA(h - 1)$, so DM truncate at $h - 1$ lags with a **rectangular** kernel
 - ▶ the rectangular estimate can be **negative**; Newey–West (Bartlett) weights keep it positive [Newey and West \(1987\)](#), at the cost of downward bias
 - ▶ suboptimal forecasts have errors with longer memory: use a data-driven bandwidth ([Chapter 0](#))
- **HLN correction** [Harvey, Leybourne and Newbold \(1997\)](#): $DM^* = DM \sqrt{(P + 1 - 2h + h(h - 1)/P)/P}$ compared with t_{P-1}
 - ▶ the factor removes the bias of the variance estimator for $MA(h - 1)$ errors; the t distribution handles small P

Size of DM and HLN in small samples



Two independent $MA(h-1)$ error series with equal variance, squared loss; share of rejections at 5% in 4 000 replications

Interpreting the size experiment

- $h = 1$: both tests are close to 5%; DM over-rejects only at $P = 16$ (7.4%)
- $h = 8, P = 16$: DM rejects a true null 43.6% of the time, HLN 14.6%; at $P = 64$: 13.2% and 8.1%
 - ▶ the HAC variance with few observations and many lags is noisy and biased down
- Rule: at multi-step horizons with fewer than about 100 forecasts, use HLN and treat p -values near 5% as inconclusive; fixed- b critical values ([Chapter 0](#)) are an alternative

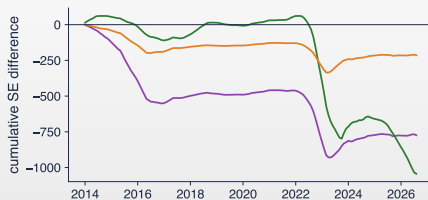
Romanian inflation one year ahead: the design

- ▣ Target: HICP inflation, y/y, 12 months ahead; forecast origins from January 2013, targets January 2014 to August 2026, $P = 152$ forecasts
 - ▶ HICP inflation in August 2026: 6.3%
- ▣ Forecasts
 - ▶ **no change**: $\hat{\pi}_{t+12} = \pi_t$; **direct AR(3)**: π_{t+12} on $1, \pi_t, \pi_{t-1}, \pi_{t-2}$, OLS on a 10-year rolling window
 - ▶ **target**: the BNR inflation target, 2.5%; **average** of AR(3) and no change
- ▣ Overlapping 12-month targets: errors are serially correlated up to lag 11; DM with $h = 12$

Romanian inflation forecasts one year ahead



— HICP inflation, Romania (y/y, %)
 — no change (made 12 months earlier)
 — direct AR(3), rolling 10 years
 - - - BNR target 2.5%



— AR(3) against no change
 — target against no change
 — average of AR and no change against no change

- Left: inflation and the forecasts made 12 months earlier; right: cumulative sum of $e_{nc,t}^2 - e_{m,t}^2$ (falling: model m loses to no change)

Interpreting the Romanian inflation comparison

Forecast	RMSE	DM	HLN	p (HLN)	p , abs. loss
no change	3.57	—	—	—	—
AR(3)	4.23	1.43	1.32	0.188	0.381
BNR target	4.43	1.30	1.21	0.230	0.319
average	3.76	0.96	0.88	0.378	0.795

- Every model has a higher RMSE than no change, yet no difference is significant: with $P = 152$ overlapping targets the effective sample is small
- Newey–West instead of the rectangular kernel: HLN 1.47 for AR(3); the conclusion does not depend on the kernel
- Relative RMSE of AR(3): 1.44 before July 2021, 1.10 after: the ranking is not stable over time [Giacomini and Rossi \(2010\)](#)

Conditional predictive ability: Giacomini–White

- $H_0: \mathbb{E}[d_{t+h} | \mathcal{G}_t] = 0$, i.e. no instrument $h_t \in \mathcal{G}_t$ predicts which method wins [Giacomini and White \(2006\)](#)
 - ▶ statistic $GW = P \bar{Z}' \hat{\Omega}^{-1} \bar{Z} \rightarrow \chi^2(q)$, $Z_t = h_t d_{t+h}$, $\hat{\Omega}$ HAC with $h - 1$ lags
 - ▶ $h_t = 1$ gives the unconditional test; adding d_t or a state variable tests whether the past predicts the winner
- Valid with a **rolling** window of fixed size, nested or non-nested models, any loss; invalid with a recursive window
 - ▶ a rejection suggests a decision rule: use model 1 when $\hat{\delta}' h_t > 0$
- Romanian inflation, AR(3) against no change, $h_t = (1, d_t, \pi_t - 2.5)$: $GW = 3.19$, p 0.363: neither the recent loss nor the distance from the target predicts the winner

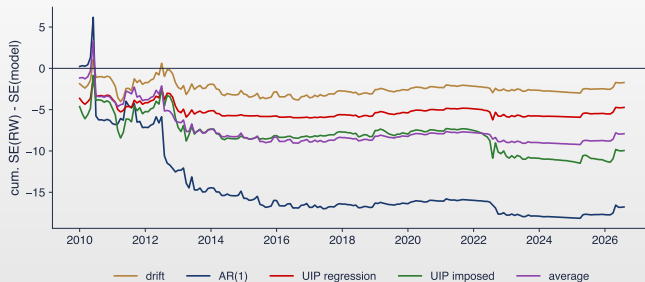
Nested models: why DM fails and Clark–West

- Model 1 (small, e.g. the random walk) is nested in model 2; under H_0 the extra coefficients are zero
 - ▶ population MSPEs are equal, and d_t is degenerate: the DM statistic is not asymptotically Normal [Clark and McCracken \(2001\)](#)
 - ▶ in finite samples model 2 estimates zero coefficients with noise, so its MSPE is **larger**: DM is undersized and has little power
- **Clark–West** [Clark and West \(2007\)](#): adjust the larger model for the noise it adds
 - ▶ $f_t = e_{1t}^2 - [e_{2t}^2 - (\hat{y}_{1t} - \hat{y}_{2t})^2]$; one-sided t test of $\mathbb{E}f_t = 0$ against $\mathbb{E}f_t > 0$ with Normal critical values
 - ▶ under H_0 , $\mathbb{E}e_1^2 - \mathbb{E}e_2^2 = -\mathbb{E}(\hat{y}_1 - \hat{y}_2)^2$: the adjustment recentres the differential ([Appendix](#))
 - ▶ approximately Normal for rolling and recursive schemes; the exact critical values of [Clark and McCracken \(2001\)](#) are non-standard

Case study: Meese and Rogoff on EUR/RON

- ▣ Meese and Rogoff (1983): structural exchange-rate models do not beat the random walk out of sample at horizons up to one year
 - ▶ their design: rolling out-of-sample forecasts, RMSE against the random walk; the result has survived forty years
- ▣ Our data: monthly EUR/RON (BNR reference rate, end of month), $\Delta s_{t+1} = 100 \Delta \ln S_{t+1}$; recursive estimation, forecasts January 2010 to August 2026 ($P = 200$)
 - ▶ nested in the random walk ($\Delta \hat{s} = 0$): drift, $AR(p)$, the UIP regression $\Delta s_{t+1} = a + b(i_t^{RO} - i_t^{EA})/12 + u$ Fama (1984)
 - ▶ no estimated parameters: UIP imposed ($a = 0, b = 1$), momentum (mean of the last three changes)
- ▣ UIP: uncovered interest parity, the expected depreciation equals the interest differential

EUR/RON: models against the random walk



- Cumulative $\sum_t (e_{RW,t}^2 - e_{m,t}^2)$ Welch and Goyal (2008): a line below zero means model m has accumulated more squared error than the random walk

Interpreting the EUR/RON tests

Model	RMSE / RW	DM (HLN)	CW	p (CW, one-sided)
drift	1.006	0.27	1.06	0.145
AR(1)	1.055	1.13	-0.15	0.561
AR(4)	1.062	1.29	-0.43	0.666
UIP regression	1.016	0.76	0.10	0.458
UIP imposed	1.033	0.99	—	—
momentum	1.183	2.61	—	—
average	1.026	0.85	—	—

- Every model has a higher RMSE than the random walk (random-walk RMSE 0.86 pp a month); the Meese–Rogoff result holds for the leu
- Clark–West does not reject either: the adjustment is not enough to make AR or UIP useful; only the drift gets close
- Managed float: the BNR smooths EUR/RON, so monthly changes are small and close to unpredictable from these predictors

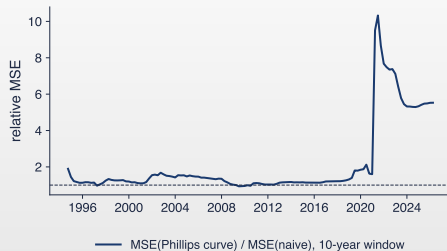
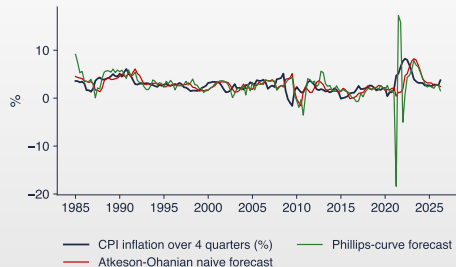
Rationality: Mincer–Zarnowitz and encompassing

- **Mincer–Zarnowitz** Mincer and Zarnowitz (1969): $y_{t+h} = a + b\hat{y}_{t+h|t} + u_{t+h}$; under squared loss an optimal forecast has $(a, b) = (0, 1)$
 - ▶ Wald test with HAC errors (u is $MA(h-1)$); $b < 1$: forecasts too volatile; $b > 1$: too timid
 - ▶ extended version: add variables in $\mathcal{F}_t$; a significant coefficient means unused information
- **Encompassing** Chong and Hendry (1986), Harvey, Leybourne and Newbold (1998): forecast 1 encompasses forecast 2 if 2 adds nothing to 1
 - ▶ $y = (1 - \lambda)\hat{y}_1 + \lambda\hat{y}_2 + \varepsilon$, $H_0: \lambda = 0$; HLN test: DM–HLN on $d_t = e_{1t}(e_{1t} - e_{2t})$, one-sided
 - ▶ if neither encompasses the other, a combination beats both (Section 5)
- MZ assumes squared loss: under asymmetric loss a rational forecast is biased (Section 1)

Case study: Atkeson and Ohanian (2001)

- Atkeson and Ohanian (2001): a naive forecast, “inflation over the next four quarters equals inflation over the last four quarters”, beats Phillips-curve forecasts in the US after 1984
 - ▶ the Phillips curve of Stock and Watson (1999): $\pi_{t+4}^4 - \pi_t^4 = a + \beta(L)u_t + \gamma(L)\Delta\pi_t + e_{t+4}$, unemployment u_t , quarterly inflation π_t
- Our update: CPI (quarterly average) and the unemployment rate (FRED); four lags of each, rolling window of 15 years; targets 1985Q1 to 2026Q2 ($P = 166$)
 - ▶ October 2025 CPI was not published; the 2025Q4 average uses two months
- Tests: DM–HLN with $h = 4$; GW with $h_t = (1, d_t)$ and $h_t = (1, u_t)$; encompassing in both directions

The naive forecast against the Phillips curve



- Left: four-quarter CPI inflation and the two forecasts made four quarters earlier; right: MSE ratio on a moving 10-year window

Interpreting the Atkeson–Ohanian update

- RMSE: naive 1.69, Phillips curve 2.97 (ratio 1.75); by period: 1.29 (1985–2007), 1.10 (2008–2019), 2.38 (2020–2026)
- Yet DM–HLN = 1.16, p 0.248: the large post-2020 errors inflate the HAC variance as much as the mean
 - ▶ the moving ratio peaks at 10.3 in 2021Q3: the Phillips curve extrapolated the pandemic unemployment spike
- GW: p 0.355 with the lagged differential, p 0.322 with unemployment; encompassing: H_0 “naive encompasses PC” p 0.289; “PC encompasses naive” p 0.118
- The AO conclusion holds in point estimates, and a test on 41 years of overlapping data cannot separate the two

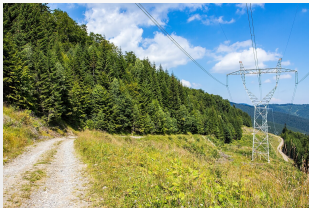
Many models: data snooping, the reality check and SPA

- ▣ Testing k models against a benchmark and reporting the best p -value is **data snooping**
 - ▶ $H_0: \max_{j \leq k} \mathbb{E}[d_{j,t}] \leq 0$ with $d_{j,t} = L_{0,t} - L_{j,t}$: no model beats the benchmark
- ▣ **Reality check** [White \(2000\)](#): statistic $\max_j \sqrt{P} \bar{d}_j$, p -value from the stationary bootstrap ([Chapter 0](#)) under the least favourable configuration
 - ▶ conservative: very poor models inflate the null distribution
- ▣ **SPA** [Hansen \(2005\)](#): studentise each $\bar{d}_j$ and drop clearly inferior models from the null distribution (consistent p -value)
 - ▶ lower and upper p -values bracket it; the upper one equals the RC with studentisation
 - ▶ StepM [Romano and Wolf \(2005\)](#) identifies *which* models beat the benchmark
- ▣ EUR/RON, $k = 9$ models against the random walk: SPA $p = 0.53$ (lower), 0.85 (consistent), 0.88 (upper): no model beats it

The model confidence set

- ▣ **MCS Hansen, Lunde and Nason (2011)**: the set $\hat{\mathcal{M}}_{1-\alpha}$ that contains the best model(s) with probability at least $1 - \alpha$ asymptotically
 - ▶ no benchmark: all models are treated symmetrically; analogous to a confidence interval for a parameter
- ▣ Algorithm: start with all m models
 - ▶ **equivalence test**: $H_0: \mathbb{E}[d_{i\cdot,t}] = 0$ for all i , $d_{i\cdot,t} = L_{i,t} - \frac{1}{m} \sum_j L_{j,t}$, statistic $T_{\max} = \max_i \bar{d}_{i\cdot} / \widehat{\text{se}}(\bar{d}_{i\cdot})$
 - ▶ if rejected, **eliminate** the model with the largest t_i and repeat; stop at the first non-rejection
 - ▶ block bootstrap for the null distribution and the standard errors; MCS p -value of a model = the largest test p -value up to its elimination
- ▣ A large MCS is a finding too: the data cannot separate the models (a weak sample, not equal models)

Case study: day-ahead electricity load in Romania

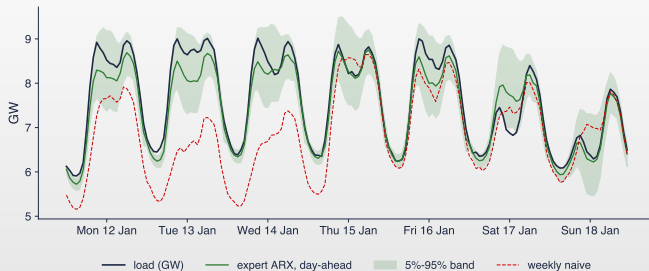


A 400 kV line, Predeluș Pass (2018)

Photo: Drprv (2018); CC BY-SA 3.0; Wikimedia Commons

- Target: the 24 hourly loads of day $d + 1$ with data up to day d ; 1 January 2025 to 30 September 2026 (638 days); mean 5.98 GW
 - ▶ quantile forecasts at the 99 percentiles, scored by the average pinball loss as in GEFCom2014 Hong et al. (2016), Hong and Fan (2016)
- Models: naive day, weekly naive, mean of 4 weeks; the expert ARX of Ziel and Weron (2018) for each hour
 - ▶ regressors: $y_{d,h}$, $y_{d-1,h}$, $y_{d-6,h}$, min and max of day d , last hour of d , Monday, Saturday, Sunday; one-year rolling window
 - ▶ quantiles: point forecast plus the empirical quantiles of the last 182 errors at the same hour
- ARX: autoregression with exogenous variables (here calendar dummies)

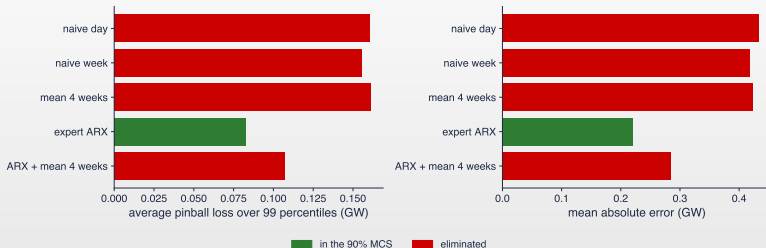
A winter week of load forecasts



- 12–18 January 2026; the weekly naive forecast copies the holiday week of 5–9 January (Orthodox Epiphany and St John, 6–7 January)

Q ATS_ch1_load_mcs

Average pinball loss and the 90% MCS



□ Daily average losses; MCS with $T_{\max}$, moving-block bootstrap (blocks of 7 days), 2000 replications; green: in the MCS

Q ATS_ch1_load_mcs

Interpreting the load comparison

Model	Pinball (MW)	MAE (MW)	90% coverage	MCS p
naive (day d)	161	433	89.2%	< 0.001
weekly naive	155	417	87.6%	< 0.001
mean of 4 weeks	161	423	88.7%	< 0.001
expert ARX	83	221	88.2%	1.000
ARX + mean of 4 weeks	107	285	88.3%	< 0.001

- The expert ARX halves the losses of the naive rules and is alone in the MCS; DM–HLN against the ARX + mean combination: -10.7
- Here combining with a much worse forecast hurts: equal weights help only when the forecasts are of similar quality (Section 5)
- All intervals under-cover slightly; the largest losses fall on and right after public holidays (8 January 2026, 25 December 2025, 1 January 2025): calendar effects are the next improvement

Recap: Comparing forecasts

- DM compares forecasts; with estimated models choose West (population) or GW (methods, rolling window)
- Multi-step, small P : HAC variance plus the HLN correction and t_{P-1}
- Nested models: Clark–West; many models: SPA or the MCS, never the best p -value
- Romanian inflation, US inflation, EUR/RON: the simple benchmark is hard to beat and differences are rarely significant

Bates and Granger (1969)



Clive W. J. Granger (1934–2009), Nobel
Memorial Prize 2003

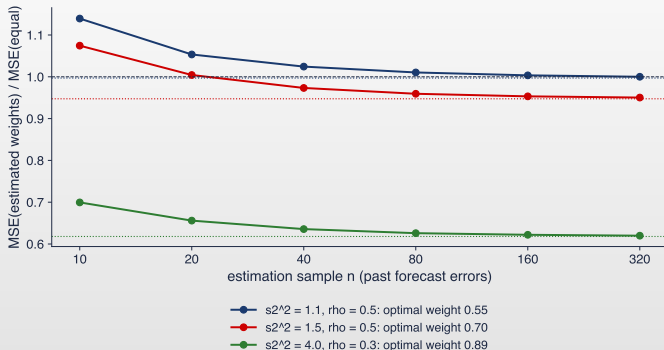
Photo: Olaf Storbeck (2008); CC BY-SA 2.0;
Wikimedia Commons

- Two unbiased forecasts with error variances σ_1^2, σ_2^2 and covariance σ_{12} ; combination $w\hat{y}_1 + (1 - w)\hat{y}_2$ [Bates and Granger \(1969\)](#)
 - ▶ error variance $w^2\sigma_1^2 + (1 - w)^2\sigma_2^2 + 2w(1 - w)\sigma_{12}$, minimised at
 - ▶ $w^* = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}, \quad \sigma_c^2 = \frac{\sigma_1^2\sigma_2^2 - \sigma_{12}^2}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} \leq \min(\sigma_1^2, \sigma_2^2)$
- N forecasts: $w^* = \Sigma^{-1}\iota/(\iota'\Sigma^{-1}\iota)$; [Granger and Ramanathan \(1984\)](#): regress y on the forecasts, with or without a constant and the sum-to-one restriction
 - ▶ combination diversifies model risk, structural breaks and the private information of forecasters [Timmermann \(2006\)](#)

The forecast combination puzzle

- Empirical regularity: the simple average beats the “optimal” estimated weights [Stock and Watson \(2004\)](#), [Timmermann \(2006\)](#), [Genre et al. \(2013\)](#)
 - ▶ Stock–Watson: across 7 countries and many predictors of output growth, equal weights and the median are among the best combinations
- **Explanation** [Smith and Wallis \(2009\)](#), [Claeskens et al. \(2016\)](#): the weights are estimated
 - ▶ $\mathbb{E}[\text{MSE}(\hat{w})] = \text{MSE}(w^*) + \mathbb{E}(\hat{w} - w^*)^2 D$, $D = \sigma_1^2 + \sigma_2^2 - 2\sigma_{12}$
 - ▶ gain of w^* over $1/2$: $(w^* - 1/2)^2 D$; loss from estimating: $\text{Var}(\hat{w})D$, of order $1/n$
 - ▶ equal weights win when $(w^* - 1/2)^2 < \text{Var}(\hat{w})$: similar forecasts, short samples, correlated errors
- Also: weights move with breaks, so long windows are biased and short ones noisy

The puzzle by simulation



- Bivariate Normal errors with $\sigma_1^2 = 1$; weights estimated from n past errors; 4 000 replications; dotted: population ratio $\text{MSE}(w^*) / \text{MSE}(1/2)$

Interpreting the puzzle simulation

- Similar forecasts ($w^* = 0.55$): the optimal weight lowers the population MSE only to 0.997 of the equal-weight MSE, while estimating it costs: ratio 1.14 at $n = 10$, 1.024 at $n = 40$
- Moderately different ($w^* = 0.70$): estimated weights need about 20 past errors to break even (1.07 at $n = 10$, 0.973 at $n = 40$)
- Very different ($w^* = 0.89$): estimate the weights, or drop the weak forecast; the load case of Section 4 is of this kind

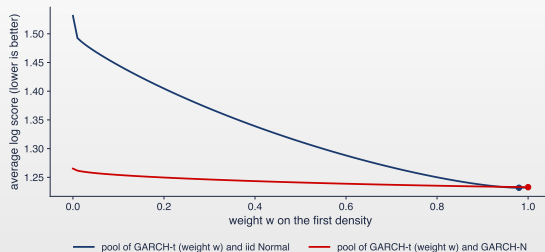
Between equal and optimal: shrinkage and robust combinations

- **Shrinkage:** $\hat{w}_\lambda = \lambda \hat{w} + (1 - \lambda)\iota/N$ Diebold and Pauly (1990), Stock and Watson (2004)
 - ▶ a Bayesian prior centred on equal weights; λ falls with the number of forecasts and rises with the sample
- **Performance weights:** $w_i \propto 1/\text{MSE}_i$ on a recent window, ignoring correlations; discounting gives recent errors more weight
 - ▶ **previous best:** select the forecaster with the lowest past MSE (a weight of one)
- **Robust:** median, trimmed mean: protect against outlying forecasters
- Panels of experts change: forecasters enter and exit, so Σ is never fully observed Capistrán and Timmermann (2009)
 - ▶ performance weights need a minimum record; a review of 50 years of practice: Wang et al. (2023)

Combining density forecasts

- **Linear pool:** $p(y) = \sum_i w_i p_i(y)$, $w_i \geq 0$, $\sum w_i = 1$ Hall and Mitchell (2007)
 - ▶ weights that minimise the average log score: **optimal prediction pools** Geweke and Amisano (2011)
 - ▶ interior weights even when every model is misspecified: a pool is not a model average in the Bayesian sense
- The linear pool of calibrated densities is **over-dispersed** Gneiting and Ranjan (2013)
 - ▶ the mixture adds the variance of the means; remedy: a beta-transformed pool or quantile averaging (Vincentisation)
- Proper scores (Section 3) are the natural criteria for estimating and comparing pools

Optimal pools for the S&P 500 densities



- Average log score of $w p_{\text{GARCH-}t} + (1 - w) p_2$ on 2013–2026; dots: minima
- With the i.i.d. Normal: $w^* = 0.98$, score 1.2318 against 1.2328 for GARCH- t alone; with GARCH-N: $w^* = 1.00$: GARCH- t dominates, the pool adds almost nothing

Interpreting the optimal pools

- The pool with the i.i.d. Normal puts almost all weight on GARCH- t : a small Normal component only insures against extreme days
- With GARCH-N the optimum is the corner $w = 1$: the two GARCH densities carry the same information, the t tails are simply better
- Weights estimated on the evaluation sample are optimistic; in practice estimate them on a past window and evaluate on the next

Case study: can anything beat the simple average?

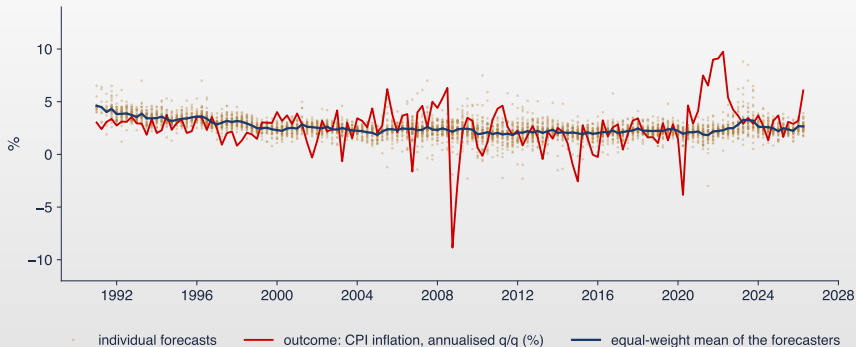


Federal Reserve Bank of Philadelphia, home of the SPF since 1990

Photo: Beyond My Ken (2013); CC BY-SA 4.0;
[Wikimedia Commons](#)

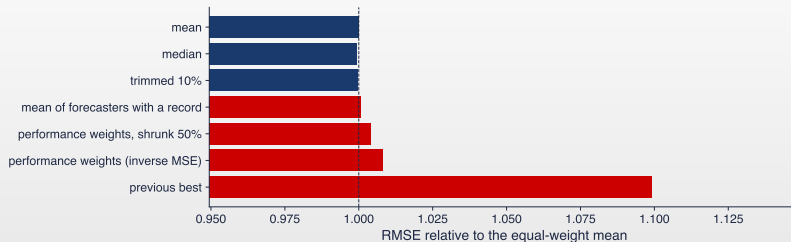
- Genre et al. (2013) compare combination schemes on the ECB Survey of Professional Forecasters: equal weights, median, trimmed mean, performance-based weights, shrinkage, and others
 - ▶ their answer: rarely, and not robustly over time
- We apply their schemes to the US SPF: CPI inflation (annualised q/q) four quarters after the survey quarter; surveys 1990Q1–2025Q2 ($P = 142, 195$ forecasters, on average 36 per survey)
 - ▶ performance weights: inverse MSE over the last 20 surveys with a known outcome, at least 8 past forecasts (on average 25 eligible); trimmed mean: 10% on each side

Individual SPF forecasts and the outcome



- Each dot: one forecaster's CPI forecast for the quarter four quarters ahead, placed at the target quarter; the outcome is far more volatile than any forecast

Combination schemes against the mean



□ RMSE relative to the equal-weight mean (RMSE 2.18 pp); blue: at least as good as the mean

Q.ATS_ch1_spf_combination

Interpreting the SPF combination

- Median 0.999, trimmed mean 1.000, inverse-MSE 1.008, previous best 1.099 (relative RMSE)
 - ▶ DM–HLN against the mean: median p 0.842; inverse MSE p 0.091; previous best HLN 2.76, p 0.007
 - ▶ Genre et al.'s answer holds on US data: nothing beats the average significantly, and chasing the past winner loses
- Yet 33% of the 76 forecasters with at least 20 forecasts have a lower RMSE than the mean
 - ▶ they are known only ex post, over different and shorter samples: survivorship and luck, not skill one could have selected
- Mincer–Zarnowitz on the mean: $\hat{b} = 0.24$ (SE 0.31), $R^2 = 0.005$, $p < 0.001$: one-year-ahead quarterly CPI inflation is almost unpredictable; the forecasts are smooth, the outcomes noisy

Lessons from the M4 and M5 competitions

- **M4** (2018) [Makridakis et al. \(2020\)](#): 100,000 series, 61 methods, point forecasts and 95% intervals
 - ▶ the winner was a hybrid of exponential smoothing and a recurrent neural network; the second, a combination of statistical methods with weights learned by a meta-learner
 - ▶ combinations dominated the top places; pure machine learning methods did poorly; prediction intervals were too narrow for most methods
- **M5** (2020) [Makridakis et al. \(2022\)](#): 42,840 hierarchical Walmart sales series, weighted scaled errors
 - ▶ gradient-boosted trees (LightGBM) trained *across* series won; exogenous variables (prices, events) and cross-learning mattered
 - ▶ again, ensembles of many models were at the top ([Chapter 12](#))
- Common lesson: combine, evaluate with scale-free scores (MASE, [Chapter 0 of TSA](#)) and proper scores, and report uncertainty that is calibrated

Recap: Combination

- Optimal weights reduce the error variance; estimated weights add a cost of order $1/n$
- The puzzle: when forecasts are similar, equal weights, the median and the trimmed mean win
- Linear pools of densities are over-dispersed; estimate them by the log score
- US SPF: no scheme beats the mean significantly; selecting the past winner loses

Vintages and revisions

- Macroeconomic data are revised: US GDP is first published about 30 days after the quarter and revised for years (benchmark revisions, new methods)
 - ▶ a **vintage** is the data set as available on a given date; the real-time data set stores all vintages [Croushore and Stark \(2001\)](#)
- Sources without a key: ALFRED (St. Louis Fed, vintages of FRED series), the Philadelphia Fed RTDSM; the ECB keeps a real-time database for the euro area; INS publishes a flash estimate of GDP and later revised estimates
 - ▶ forecasts made with latest-vintage data overstate what was predictable in real time [Stark and Croushore \(2002\)](#), [Croushore \(2011\)](#)
- Question for the evaluation: which vintage is the “truth”? The first release (what forecasters target) or the latest (the best estimate of reality)?

US real GDP growth: first release and latest vintage



- Annualised q/q growth, 1990–2026; revisions = latest minus first release (Philadelphia Fed RTDSM); the 2025Q4 first release is missing (government shutdown)

Interpreting the revisions

- Mean revision 0.23 pp, standard deviation 1.54 pp, mean absolute revision 1.21 pp over 143 quarters; correlation first–latest 0.939
 - ▶ largest: 2020Q2 (4.9 pp) and 2008Q4 (−4.7 pp): revisions are large exactly at turning points
- SPF current-quarter nowcast ($P = 143$): RMSE 1.95 against the first release, 2.36 against the latest; MZ slope 1.15 and 1.08
- The ranking of forecasters and models can change with the vintage: state the choice in the pre-registration

A pre-registered evaluation design

- Fix **before** looking at the out-of-sample results
 - ▶ target, horizons, loss or score (Sections 1–3), vintage of the outcome, evaluation period, estimation scheme and window
 - ▶ benchmark (random walk, seasonal naive, survey mean) and the full list of competing models
 - ▶ tests: DM–HLN or GW; Clark–West if nested; SPA or MCS for many models; a correction for several horizons (Holm)
- Report everything that was tried; a model selected on the test period has a biased error
 - ▶ the project of this course starts with exactly such a plan (proposal and pre-registration, 5%)

Recap: Real-time data

- Evaluate forecasts with the data that existed when they were made
- Revisions are large at turning points; the choice of the “truth” vintage changes RMSE and rankings
- Pre-register targets, losses, benchmarks, tests and the vintage

An open question

- Does the forecast combination puzzle survive when the panel of experts changes over time and the outcome distribution shifts (2020–2023)?
 - ▶ formal: $H_0: \mathbb{E}[(y - \hat{y}_\lambda)^2 - (y - \bar{y})^2] \geq 0$ for every shrinkage factor λ fixed in advance, in every sub-period
 - ▶ falsified by a pre-registered λ that beats the mean with DM–HLN $p < 0.05$ after a Holm correction across sub-periods
- Why it matters: central banks publish the survey mean; a better combination would change the published expectations
 - ▶ literature to start from: Genre et al. (2013), Capistrán and Timmermann (2009), Smith and Wallis (2009), Claeskens et al. (2016), Wang et al. (2023)

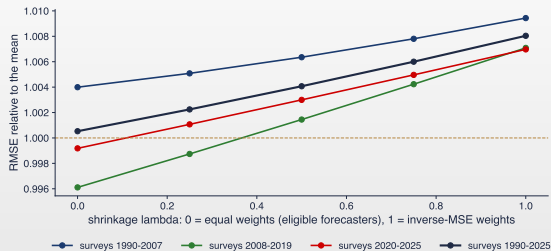
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
 - ▶ **literature:** List peer-reviewed papers on combining SPF forecasts with entry and exit of forecasters; give DOIs. Then check every DOI on Crossref
 - ▶ **hypothesis:** Propose three reasons why inverse-MSE weights could beat the mean after 2020, each with a test.
 - ▶ **code and replication:** ask for a function, then reproduce a known number first (the relative RMSE of the median in this lecture)
 - ▶ **robustness and critique:** Act as a hostile referee: list every way this evaluation could be data-snooped.
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

What the human checks

- Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- No look-ahead: weights at survey s use only outcomes known at s (here, forecasts from at least five surveys earlier, whose target quarter has ended)
- The HAC lag matches the overlap of the targets ($h - 1$); the HLN correction is applied
- λ , sub-periods and the minimum record are fixed before the results; all variants are reported
- The interpretation separates “not significant” from “equal”

Mini-case: shrinkage across sub-periods



- RMSE of $\lambda \cdot (\text{inverse-MSE weights}) + (1 - \lambda) \cdot (\text{equal weights among eligible forecasters})$, relative to the mean of all forecasters; US SPF, CPI four quarters ahead
- $\lambda = 1$: 1.009 (1990–2007, $P = 72$), 1.007 (2008–2019), 1.007 (2020–2025); full sample HLN 1.70, p 0.091
- Every sub-period: the more weight on past performance, the worse; the puzzle survives the 2020–2023 inflation surge

Project idea

- ▣ **Combining Romanian inflation forecasts:** models (AR, Phillips curve with the output gap, BNR target), surveys and market-based expectations
 - ▶ pre-register horizons 1–12 months, the pinball loss at 5%, 50%, 95% and the CRPS, the Holm correction across horizons
 - ▶ replicate first: the Romanian inflation table of this lecture (Seminar 1, C1 does it by horizon)
 - ▶ extension: density combination by optimal pools; GW test of whether the 2022 shock changed the winner
- ▣ Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- ▣ Choose the loss or score first: it defines what a “good” forecast is (mean, median, quantile, distribution)
- ▣ Density forecasts: check calibration with PIT, rank with proper scores; ES needs VaR to be scored
- ▣ Tests: DM–HLN with HAC, GW for methods, Clark–West for nested models, SPA and MCS for many
- ▣ Combine similar forecasts with equal weights; estimated weights pay off only with very different forecasts and long records
- ▣ Evaluate in real time and pre-register the design

Self-assessment

Questions

- ▣ Which functional does the pinball loss with $\tau = 0.05$ elicit?
- ▣ What does a U-shaped PIT histogram say about the intervals?
- ▣ Why is DM not valid for nested models with estimated parameters?
- ▣ When do estimated combination weights beat equal weights?
- ▣ Why can ES not be ranked on its own?

Next:

- ▣ Structural breaks and nonlinear models: tests for breaks, threshold models
- ▣ Forecast evaluation under instability: rolling windows, forecast breakdowns, the fluctuation test [Giacomini and Rossi \(2010\)](#)

Appendix: consistency of the pinball loss

- $\mathbb{E}\rho_\tau(Y - x) = \tau \int_x^\infty (y - x)dF(y) + (1 - \tau) \int_{-\infty}^x (x - y)dF(y)$
- Derivative (Leibniz): $-\tau(1 - F(x)) + (1 - \tau)F(x) = F(x) - \tau$; zero at $x = F^{-1}(\tau)$; second derivative $f(x) \geq 0$
- For $\tau = 1/2$: $\rho_{1/2}(e) = |e|/2$, so the median; for x off the quantile the excess loss is $\int_q^x (F(z) - \tau)dz > 0$
- Generalised piecewise linear losses $(1\{y \leq x\} - \tau)(g(x) - g(y))$, g increasing, are the whole class of consistent losses for the quantile [Gneiting \(2011\)](#)

Appendix: the kernel form of the CRPS

- Write $(F(z) - 1\{y \leq z\})^2 = F(z)^2 - 2F(z)1\{y \leq z\} + 1\{y \leq z\}$
- With $X, X' \sim F$ i.i.d.: $F(z)^2 = P(X \leq z, X' \leq z)$ and
 $\mathbb{E}|X - y| = \int [F(z)(1 - 1\{y \leq z\}) + (1 - F(z))1\{y \leq z\}] dz$
- $\frac{1}{2}\mathbb{E}|X - X'| = \int F(z)(1 - F(z)) dz$; subtracting gives $\int (F(z) - 1\{y \leq z\})^2 dz$
- Propriety: $\mathbb{E}_G \text{CRPS}(F, Y) - \mathbb{E}_G \text{CRPS}(G, Y) = \int (F(z) - G(z))^2 dz \geq 0$, zero only if $F = G$

Appendix: the Clark–West adjustment

- Under H_0 the larger model's extra coefficients $\beta_2 = 0$; its forecast is $\hat{y}_2 = \hat{y}_1 + \hat{\beta}_2'x_2$ approximately, with $\hat{\beta}_2$ pure estimation noise
- $e_2 = e_1 - (\hat{y}_2 - \hat{y}_1)$, so $e_1^2 - e_2^2 = 2e_1(\hat{y}_2 - \hat{y}_1) - (\hat{y}_2 - \hat{y}_1)^2$
- $\mathbb{E}[e_1(\hat{y}_2 - \hat{y}_1)] = 0$ under H_0 (e_1 is a martingale difference), hence $\mathbb{E}(e_1^2 - e_2^2) = -\mathbb{E}(\hat{y}_2 - \hat{y}_1)^2 < 0$
- Adding $(\hat{y}_1 - \hat{y}_2)^2$ recentres the differential at zero: $f_t = 2e_{1t}(\hat{y}_{2t} - \hat{y}_{1t})$, the encompassing statistic of [Harvey, Leybourne and Newbold \(1998\)](#) in disguise

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